HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · POWER TRANSMISSION · FINANCE

RECEIVABLES RECONCILIATION · POWER QUERY AUTOMATION

1,000 PDFs, One Table

Disbursement advices arrived as more than a thousand PDF and Excel files, each slightly different. HMC built a Power Query process with custom functions that reads every one, cleans it and stacks it into one table, and keeps working when the format changes.

~~3 wks~~ → **5** minutes to refresh the full dataset

1,000 + files
Advice files consolidated
PDF and Excel, DA 3011–4001

5 minutes
To refresh the full dataset
▼ from about 3 weeks

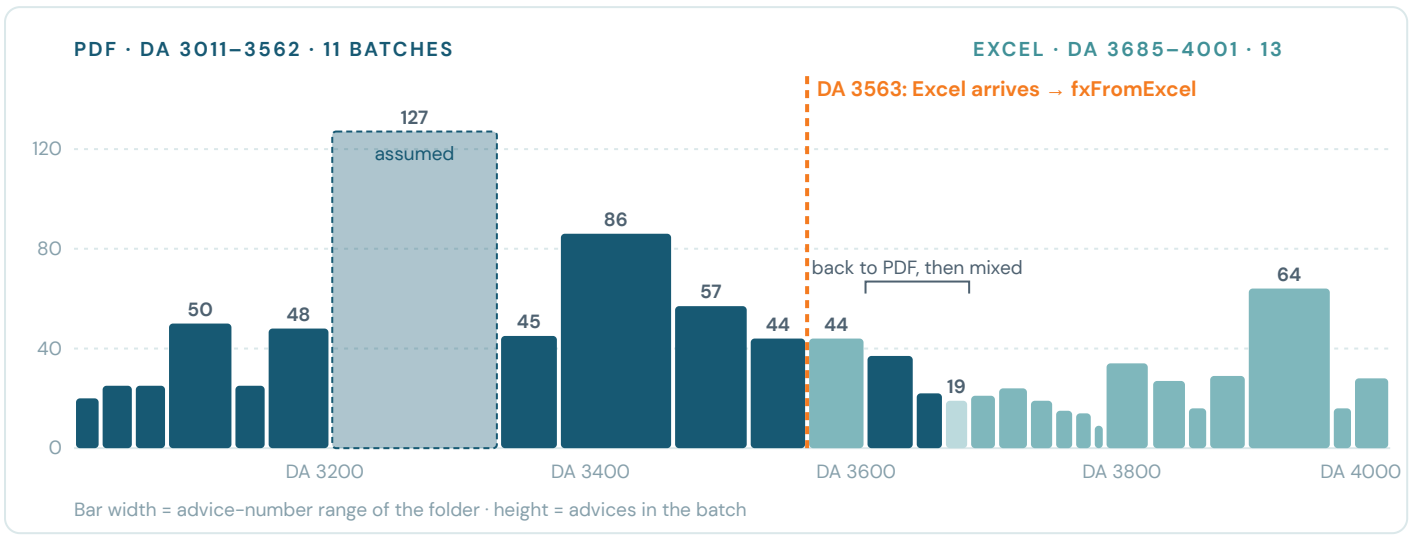
6 functions
Custom M functions
handle every file variation

0 rows
Figures re-keyed by hand
every row traced to its file

Rings: advice range by batch format (PDF white, Excel teal, mixed light); ~3 weeks of copy-paste outside vs a 5-minute refresh as the orange dot inside; one segment per custom function, the file router in orange; every row tied to its source file.

01 The format changed halfway through the series

28 batch folders, DA 3011–4001 · from the original folder names



BREAK POINT
Excel from DA 3563
Later batches arrived as Excel, with title rows, serial dates and renamed licensees. A fixed template would break.

THEN IT DRIFTED BACK
PDF again, then both
Two batches went back to PDF; DA 3666–3684 mixed PDF and Excel in one folder.

THE FIX
One router for every file
fxTransformDA reads one file and routes it to PDF or Excel handling, so every format lands in the same table.

02 One advice, and everything wrong with it

Advice No. 3687 (02.08.2016), page 1 of 3 · as delivered

Powergrid Corporation Of India Limited
(Corporate Commercial Department)

Advice No:3687 Advice Date:02.08.2016 Created By :60001593

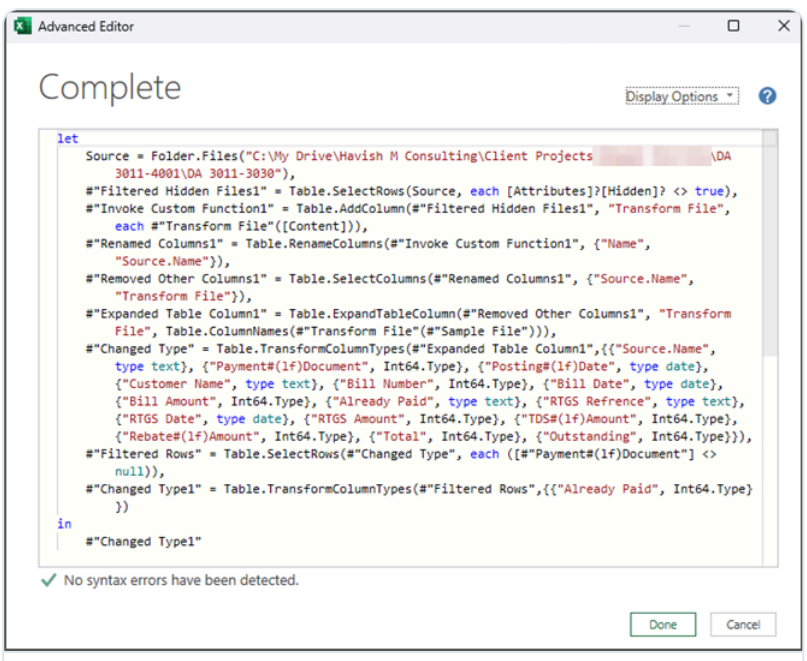
Disbursement Advice (Amount in INR)

Payment Document	Posting Date	Customer Name	Bill Number	Bill Date	Bill Amount	Already Paid	RTGS Refrence	RTGS Date	RTGS Amount	TDS Amount	Rebate Amount	Total	Outstanding
1800001156	02-08-2016	Jaipur Vidyut Vitran Nigam Lim	0091101876	05-04-2016	455,032,561	338,592,298	SBBO108167648698- 8468	01-08-2016	51,600,000	0	0	51,600,000	64,840,263
							TOTAL		51,600,000	0	0	51,600,000	

Licensee wise Payment Details

S.No	ISTS Licensees	Bill Type	Bill Raised in Month	MTC/Billing Amount in Cr.	Disbursement Ratio	Gross Realisation	Rebate Availed	TDS by DIC	Net Disbursement
1	POWERGRID	BILL-1	APR 2016	1630.23	87.1967	44,993,498	0	0	4,49,93,498
2	POWERLINKS	BILL-1	APR 2016	22.22	1.1885	613,266	0	0	6,13,266
3	WRTM	BILL-1	APR 2016	11.75	0.6285	324,306	0	0	3,24,306
4	UPPTCL	BILL-1	APR 2016	1.30	0.0695	35,862	0	0	35862
5	RRVPNL	BILL-1	APR 2016	3.12	0.1669	86,120	0	0	86120
6	PDD J&K	BILL-1	APR 2016	0.00	0.0000	0	0	0	00000
7	GETCO	BILL-1	APR 2016	0.48	0.0257	13,261	0	0	13261
8	MSETCL	BILL-1	APR 2016	8.14	0.4354	224,666	0	0	2,24,666
9	WRTG	BILL-1	APR 2016	3.71	0.1984	102,374	0	0	1,02,374
10	JAYPEE POWERGRID	BILL-1	APR 2016	15.13	0.8093	417,599	0	0	4,17,599

Page 1 of 3



1. **Wrapped headers**: "Bill Raised in↵Month" → Bill Month
2. **RTGS suffix**: "SBBO108167648698- 8468" split to the UTR
3. **Total rows** sitting inside the tables, removed
4. **Indian commas**: 4,49,93,498 parsed to 44,993,498
5. **Zeros as text**: "00000" and "–" become 0
6. **Page footers** ("Page 1 of 3") dropped

1. **The folder query.** Folder.Files → filter hidden files → invoke the Transform File function → expand → set types. One folder name is pixelated.

CHECK BUILT INTO THE LOAD
Ratio × RTGS amount = gross realisation
87.1967% × 5,16,00,000 = 4,49,93,498
Row 1 of this advice ties. Every row is checked the same way on refresh; any row that fails is listed with its source file name.

03 Name every variation, give it a function

Variability catalogue → small, reusable M functions

fxCleanHeaders
Wrapped headers
~~Bill Raised in↵Month~~
→ Bill Month

fxCleanHeaders
Header synonyms
~~Net↵Disbursement (Rs.)~~
→ Net Disbursement ₹

Table.SelectRows
Totals and page footers
~~TOTAL · Page 1 of 3~~
→ removed

fxParseINR
Indian vs Western commas
~~4,49,93,498~~
→ 44,993,498 (number)

fxParseINR
Zeros written as text
~~00000 / –~~
→ 0

fxMapLicensee
Licensee name variants
~~W.R.T.M. / WRTM~~
→ Western Region Transmission (Maharashtra)

fxSplitRTGS
Reference with suffix
~~SBBO108167648698– 8468~~
→ SBBO108167648698

fxFromExcel
Excel serial dates, title rows
~~42461 · 3 title rows~~
→ Apr-2016 · removed

Every file, whatever its format, ends in the same twelve columns: advice number, date, payer, bill month, licensee, UTR, billing, ratio, gross realisation, rebate, TDS and net disbursement.

04 Four pain points, four fixes

How the advice data reaches the reconciliation · before → after

BEFORE
1,000+ files
Open each PDF, copy each table: about three weeks for the full set.
FOLDER QUERY
AFTER
Drop files, press Refresh
Every file is read, cleaned and checked in about five minutes.

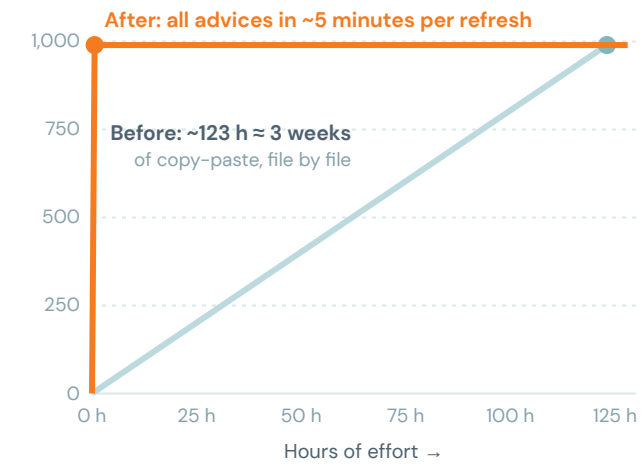
BEFORE
Unstructured data
Wrapped headers, footers and totals in tables, mixed Indian and Western commas.
FXCLEANHEADERS · FXPARSEINR
AFTER
Clean numbers, one name list
Amounts parsed to numbers; licensees mapped to one list the team edits.

BEFORE
The format kept changing
Templates broke whenever a batch arrived in a new layout; rework each time.
FXTRANSFORMDA ROUTER
AFTER
One new rule, not a rebuild
Custom functions route and clean each format; new folders picked up automatically.

BEFORE
Spot checks, no trail
Checks at month end; hard to trace a figure back to its file.
VALIDATION IN THE QUERY
AFTER
Every row checked and traced
Ratio × RTGS = gross on every row, each carrying its source file and advice number.

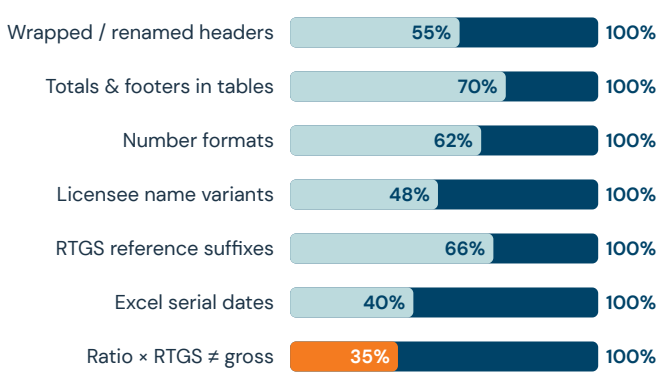
05 Building the full table

Advices consolidated vs hours of effort, DA 3011–4001 · indicative



06 Issues caught on load

Share of each issue type caught before (bar) vs after (navy track) · indicative



07 How HMC ran it

1 week — Discover
Sampled every batch; catalogued seven kinds of variation and a target schema.

1 week — Design
One twelve-column schema for every input, plus validation rules.

2 weeks — Build
Power Query workbook with six custom functions and a folder query.

✓ handover — Adopt
Save advices, press Refresh; one-page guide and walkthrough.

∞ ongoing — Scale
Mapping tables kept by finance; output feeds receivables, TDS and rebate reconciliations.

What any finance team can borrow

01 — Solve one file, then the folder
Once Power Query can read one advice reliably, a thousand is just a refresh.

02 — Name every variation
A catalogue of the ways files differ became a set of small functions. Each new quirk is one more rule, not a rebuild.

03 — Check as you load
Validation built into the query means reconciliation starts from data the team already trusts.

Built with **Power Query** · **M language** (six custom functions) · **Excel** · mapping tables | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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